Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Advanced Waste Management Technologies for Pyroprocessing and Molten-Salt-Reactor (MSR) Waste Streams**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (x), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	*Radioactive Waste Management*	*Nuclear Power*	*Nuclear-Based Advanced Technologies*
Concept of Proposed Technology	The core concept of the proposed technology is the development of advanced nuclear waste management solutions tailored for both pyroprocessing chloride salt wastes and molten salt reactor (MSR) chloride-based wastes through a combined framework of waste separation and engineered waste form immobilization. The approach integrates RPI's unique experimental and analytical capabilities with the Korean SME's reactor technology, industrial scale-up experience, and commercial deployment pathways. The main components of the proposed technology include: (1) selective separation and stabilization of waste radionuclides from complex salt waste streams; (2) conversion of separated waste elements into durable, high–waste-loading ceramic-metal (cermet) or halide-metal (halmet) composite waste forms; (3) evaluation of thermomechanical stability, and chemical durability under realistic repository environments; and (4) design of scalable industrial processes to enable integration with Korean pyroprocessing or MSR facilities (see Fig. 1 for overview of the proposed concept and methodology). Potential applications of this technology span multiple areas of the advanced nuclear sector. In pyroprocessing, it enables the safe, large-scale immobilization of chloride salt wastes generated by electrorefiners, supporting Korea's long-term waste management plan and fuel cycle strategy [1]. For molten salt reactors (MSRs), the technology offers a pathway to condition, separate, and immobilize used fuel salts or secondary coolant salts—currently among the most challenging nuclear waste			

	streams worldwide. Collectively, these outcomes position both the U.S. and Korea at the forefront of next-generation nuclear reactor development and fuel cycle technologies, establishing leading-edge solutions for advanced nuclear waste management and disposal. ***Figure 1. An integrated Framework*** *for Pyroprocessing and MSR Chloride Waste Management.*
Rationale for Joint R&D Project	This project builds upon extensive experience of RPI team in salt waste management with support from the DOE ARPA-E ONWARDS program [2], which developed a low-temperature solution chemistry process to efficiently recycle LiF and separate key radionuclides, reducing advanced reactor waste volumes [3-9]. An innovative concept utilizing metal halide perovskites (MHPs) as advanced waste forms has been explored, demonstrating exceptional capability to immobilize a broad spectrum of elements—including halides, alkali and alkaline-earth metals, transition metals, and lanthanides—from complex fluoride salt waste streams (Fig. 2 and Fig. 3). This approach is well suited for managing waste from both pyroprocessing of spent nuclear fuels and molten chloride salt reactors (MSRs) [9]. The joint R&D effort is essential because these waste streams pose challenges requiring both advanced scientific expertise and practical industrial deployment. RPI contributes unique capabilities in materials synthesis, waste separation, composite waste form development, thermomechanical characterization, and chemical durability, but these innovations must be aligned with the operational requirements of advanced reactors and pyroprocessing systems. Korean SMEs, in turn, bring strengths in MSR development, pyroprocessing engineering, and industrial-scale process integration—critical areas for translating laboratory discoveries into deployable waste management solutions. Collaboration bridges the gap between fundamental science and practical implementation. RPI provides scientific principles and experimental validation, while the Korean SME ensures engineering feasibility, compatibility with Korean molten salt technologies, and commercialization potential. Together, this partnership enables a comprehensive, scalable technology solution that neither side could achieve independently, positioning both nations to lead in advanced nuclear systems and next-generation waste management technologies.
Key Words	1. **Advanced Molten Salt Waste Management** – Integrated treatment of chloride salt wastes from pyroprocessing and MSR systems. 2. **High–Waste-Loading Composite Waste Forms** – Engineered cermet and halmet matrices designed for durable immobilization of complex salt wastes. 3. **Selective Radionuclide Separation and Salt Conditioning** – Targeted removal and stabilization of hazardous species from chemically complex molten salt streams.

	4. **Thermomechanical and Chemical Durability Assessment** – Comprehensive evaluation of waste-form performance under repository-relevant thermal, chemical, and radiation environments. **5. Scalable Industrial Process Integration** – Design of fabrication and processing routes compatible with Korean industrial facilities and commercial deployment.

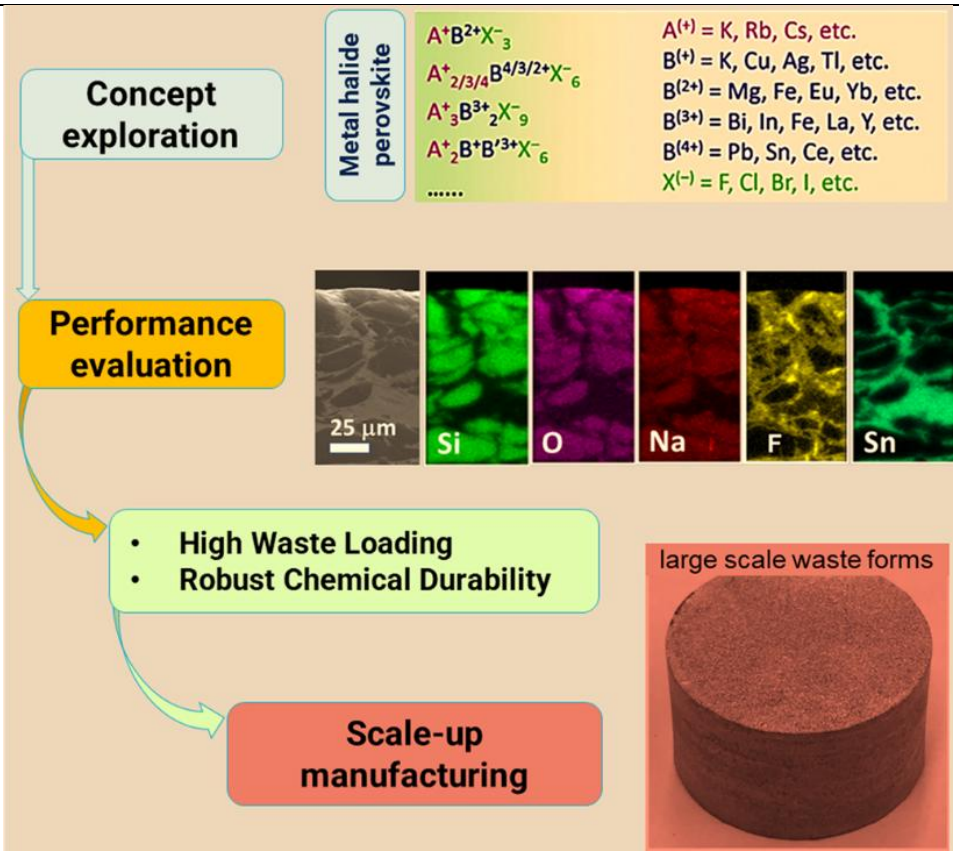

Figure 2. Key concepts of using MHPs *as innovative materials and solutions to separate and immobilize key salt elements from complex salt waste from molten fluoride/chloride reactors and pyroprocessing of spent nuclear fuels.*

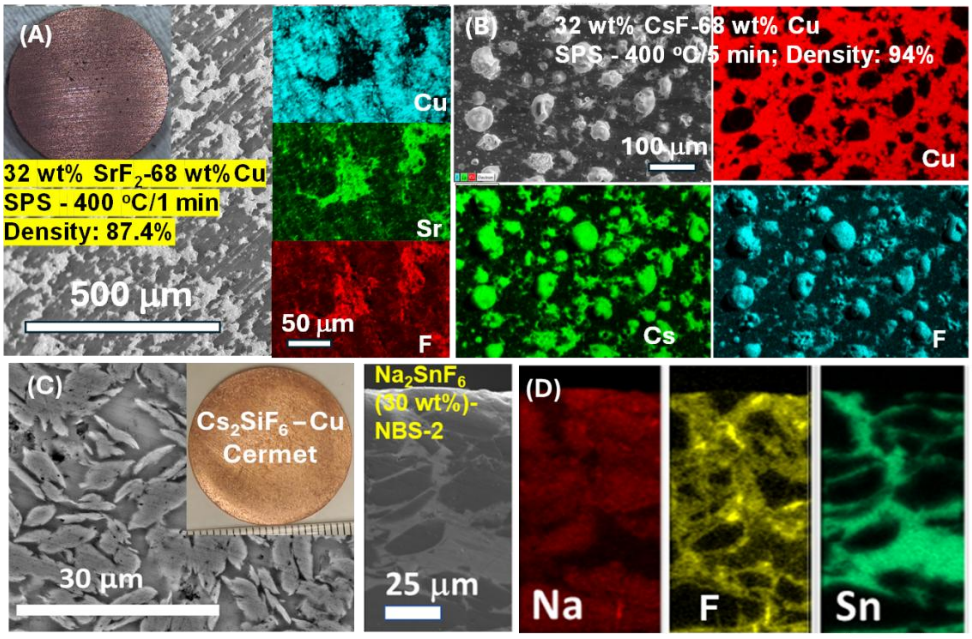

Figure 3. Proof-of-concept demonstration of halmet (A) and cermet (B) and glass-bonded MHP composite (C) waste forms *for effective immobilization of halide and salt wastes from complex fluoride salt waste streams.*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2-3)	End of the Project	Level (4-5)

R&D Objectives

The joint project aims to develop end-to-end technologies for managing molten salt wastes from pyroprocessing and MSR systems, bridging materials science with industrial implementation:

1. **Selective salt treatment and radionuclide separation:** Develop methods to separate key waste elements (halides, alkaline and alkaline-earth elements, Sr, Cs) from chemically complex chloride waste streams, reducing waste volume and optimizing composition for downstream immobilization.
2. **Engineered high–waste-loading composite waste forms:** Design and optimize ceramic–metal (cermet) and halide–metal (halmet) matrices to immobilize treated salts, achieving high waste loading while ensuring thermal stability, mechanical integrity, and low leachability.
3. **Advanced fabrication technologies:** Establish scalable production methods, including spark plasma sintering and conventional furnace sintering, to precisely control microstructure, phase stability, and interfacial bonding compatible with industrial-scale manufacturing.
4. **Performance evaluation framework:** Develop integrated thermomechanical and chemical durability assessments under repository-relevant conditions to ensure long-term stability and safety.
5. **Process scale-up and system integration:** Couple laboratory innovations with Korean SME expertise to design industrial-scale workflows, perform techno-economic assessments, and identify pathways for deployment in pyroprocessing and MSR facilities.

These objectives collectively deliver a deployable, end-to-end solution for advanced nuclear salt waste management, supporting next-generation reactor development and sustainable fuel cycle strategies.

R&D Target Specifications

Target Performance Specifications Aligned with Korean Disposal Criteria

Category	Performance Metric	Target Specification	Alignment with Korean Disposal Criteria
Salt Treatment & Conditioning	Radionuclide separation efficiency	≥95–99% for key fission products	Supports KAERI goals for hazard reduction prior to immobilization
	Removal of chemically aggressive species (e.g., Cl^-, alkali salts, Cs and Sr)	≥90% separation or stabilization	Reduces corrosion and degradation risks in disposal environments
Waste Loading & Volume Reduction	Effective salt waste loading	15–30 wt%	Higher than conventional glass, consistent with Korean's interest in compact waste forms
	Waste volume reduction factor	≥5× (relative to untreated salt disposal)	Directly supports repository footprint minimization
Waste Form Fabrication	Relative density	≥90–95% theoretical	Required for low permeability and long-term stability
Chemical Durability (Key Disposal Metric)	Normalized elemental release rate (PCT / MCC-1–type tests)	$\leq 10^{-2}$–10^{-4} $g \cdot m^{-2} \cdot day^{-1}$	Comparable to or better than reference glass/ceramic waste forms used by Korean
	Cumulative salt or radionuclide release	<1–5% over test duration	Supports long-term containment assumptions
Scalability & Deployment	Fabrication scalability	Lab → pilot scale (kg)	Aligns with Korean industry demonstration needs

R&D Plan	The proposed R&D adopts a stepwise, risk-managed development strategy that integrates salt waste conditioning, waste-form fabrication, and performance evaluation within a framework compatible with Korean disposal safety requirements. Phase 1 focuses on the development of full research plan, IP analysis and assesses technical feasibility and risk reduction, establishing whether target waste loadings, chemical durability, and process compatibility can be achieved. Phase 2 advances the most promising waste-form systems toward pilot-scale fabrication, safety-case-relevant data generation, and industrial integration with Korean SME capabilities. **Phase 1 (Months 0–6): Feasibility Exploration and Full-scale R&D Plan** **Key Tasks and Methodologies** Task 1. Establishment of full-scale R&D plans, experiments and demonstration of plans, market analysis, and IP discussion Task 2. Definition and Preparation of Representative Salt Waste Streams - Representative salt compositions will be defined in collaboration with the Korean SME based on pyroprocessing chloride salts and MSR chloride salts relevant to Korean advanced reactor programs.
	Phase 2 (Months 7–36): Demonstration, Optimization, Scale-Up, and Disposal-Relevant Validation **Objectives** • Demonstrate and optimize waste-form compositions and fabrication processes. • Generate quantitative data required for Korean safety assessments. • Demonstrate scalability and industrial relevance. • Establish pathways for deployment with Korean SME facilities. **Key Tasks and Methodologies** Task 3. Initial Waste-Form Design and Fabrication. Candidate cermet and halmet formulations will be designed to accommodate 15–30 wt% salt waste loading. Fabrication will be performed using RPI's conventional sintering and spark plasma sintering (SPS) capabilities to assess densification behavior, microstructural homogeneity, and process windows. Task 4. Waste-Form Optimization and Composition Refinement Selected waste-form systems will be systematically optimized to maximize waste loading (up to ~30 wt%) while maintaining chemical durability and microstructural stability. Parametric studies will examine the influence of salt composition, metal/ceramic fractions, and processing conditions on waste-form performance. Task 5. Advanced Chemical Durability and Repository Compatibility Testing Leaching tests under a range of pH, redox, and temperature conditions will be conducted to support long-term performance modeling. Task 6. Process Scale-Up and Industrial Integration Fabrication processes will be transitioned from laboratory scale to kg-scale batches, emphasizing reproducibility and process control. The Korean SME will lead assessments of industrial manufacturability, equipment compatibility, and cost considerations, informed by RPI's experimental data. Task 7. Developing acceptance Criteria and Deployment Planning Developing acceptance Criteria to assure waste forms meeting Korean repository requirements for chemical durability, leach resistance, structural stability, thermal stability, mechanical integrity, and comply with packaging, size, and geometry standards. Develop disposal pathway compatibility and identify commercialization or demonstration opportunities jointly with the Korean SME.

R&D Milestones and Timeline

Timeframe	Key Objective	Deliverables
Months 0–6 (Phase 1)	Establishment of full-scale R&D plans, experiments and demonstration of plans, market analysis, and IP discussion	• Technical reports of the full-scale R&D plan, market analysis and IP space.
Months 7–18 (Phase 2A)	Demonstrate and optimize designs and processes; expand performance envelope	• Demonstrate and optimized process windows; Iterative prototypes with improved performance; Refined mechanistic understanding.
Months 19–30 (Phase 2B)	Validate robustness and scale-up readiness	• Reproducibility and durability datasets; Performance validation under relevant conditions; Defined scale-up and integration pathway
Months 31–36 (Phase 2C)	Develop acceptance criteria and comply with Korean waste disposal regulation	• Final reports documenting the acceptance criteria and commercial pathway for technic transition.

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	1. **Technical Contributions:** • Perform separation of critical radionuclides, including halides and salt wastes, from complex waste streams to achieve high separation efficiency. • Develop and fabricate immobilized waste forms, including metal or ceramic matrix composite waste forms. • Characterize thermal and mechanical properties of fabricated composites. • Assess waste loading capacities and thermal stability of composites. • Evaluate chemical durability under relevant environmental conditions. 2. **Research Scope:** • Laboratory-scale development and testing of waste separation and immobilization processes. • Material design optimization to maximize waste loading and ensure stability. • Systematic characterization to link composition, structure, and performance. 3. **Resources and Capabilities:** • Separation apparatus for waste element separation and extraction. • Spark Plasma Sintering (SPS) and hot-pressing equipment for waste form consolidation. • Thermal-mechanical characterization tools including Laser Flash Analysis (LFA), Simultaneous Thermal Analysis (STA), dilatometry (DIL), and indentation testing. • Chemical analysis capability via ICP-OES and ICP-MS. 4. **Specific Tasks:** • Design and execute separation experiments for halides and salt wastes. • Fabricate composite waste forms using SPS and hot pressing and perform iterative optimization. • Conduct thermal, mechanical, and chemical durability testing using LFA, STA, DIL, indentation, and ICP-MS. • Document findings, maintain detailed records, and support project reporting and milestone reviews.
Expected R&R of Korean SME Partner	1. **Technical Contributions:** • Scale-up and industrial implementation of separation and immobilization processes. • development of industrial-scale composite waste forms. • testing and validation of performance under regulatory and operational conditions. 2**. Development Tasks:** • Pilot-scale fabrication and optimization. • integration of RPI processes into industrial workflows. • ensure compliance with Korean regulatory standards. 3. **Resources:** • Industrial facilities and pilot-scale equipment; skilled personnel; materials and consumables; technical and engineering support. 4. **Specific Tasks:** • Developing acceptance Criteria to assure waste forms meeting Korean repository requirements for chemical durability, leach resistance, structural stability, thermal stability, mechanical integrity, and comply with packaging, size, and geometry standards. • Exploring large-scale testing and validation and implement laboratory-scale findings at industrial scale.

5. Expected Outcome/Impact

Expected Outcomes and Impact	**1. Technological Achievements** • Development of efficient separation processes for critical radionuclides, including halides and salt wastes, achieving high separation efficiency. • Fabrication of advanced composite waste forms (metal and ceramic-based matrix) with optimized thermal-mechanical properties, high waste loading, and chemical durability. • Demonstration of reproducible laboratory-to-pilot scale processes suitable for industrial implementation. • Generation of a comprehensive database linking material composition, processing parameters, and performance metrics for waste form design. **2. Commercialization Potential** • Industrial-scale adoption of separation and waste form fabrication technologies by Korean SMEs and nuclear waste management industries. • Potential for licensing or collaborative development with domestic and international nuclear and materials companies. • Enhancement of domestic capability to handle and immobilize critical radionuclides, enabling technology transfer and commercialization. **3. Broader Industrial and Economic Impact** • Support of national waste management strategies and compliance with regulatory standards. • Contribution to a safer, more efficient, and sustainable nuclear waste management industry. • Strengthening the domestic industrial base in high-value materials processing and advanced composite technologies. • Potential economic benefits from commercialization, job creation, and technological leadership in critical materials handling and immobilization.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization